Critical Value vs. P-Value: Concept & Differences

1. Introduction

When performing hypothesis testing, we decide whether to **reject or fail to reject the null hypothesis** (H_0). There are **two main approaches** to make this decision:

1. **Critical Value Approach**
2. **P-Value Approach**

Both methods use the **significance level (α)** to determine statistical significance.

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2. Critical Value Approach

Concept

- The **critical value** is a **threshold** from a statistical table (Z-table, t-table, chi-square table) that defines the **rejection region**.
- If the test statistic exceeds the critical value, we **reject** H_0.

Steps in the Critical Value Method

1. **State the hypotheses** (H_0 and H_a).
2. **Choose α (significance level)** (e.g., 0.05 or 0.01).
3. **Find the critical value** using statistical tables.
4. **Calculate the test statistic** (Z, t, or chi-square).
5. **Compare test statistic with the critical value**:
   - If **test statistic** > **critical value** → **Reject** H_0
   - If **test statistic** ≤ **critical value** → **Fail to reject** H_0

Example: Critical Value Method

A company claims that the average weight of its products is **500g**. A sample of **40 items** has an average weight of **490g** with a standard deviation of **15g**. Test the claim at $\alpha = 0.05$ (two-tailed test).

Step 1: Define Hypotheses

- $H_0 : \mu = 500g$ (no difference)
- $H_a : \mu \neq 500g$ (weight is different)

Step 2: Find Critical Value

- **Two-tailed test,** $\alpha = 0.05$
- **Z-critical values = ±1.96** (from Z-table)

Step 3: Compute Z-Score

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{490 - 500}{15/\sqrt{40}}$$

$$Z = \frac{-10}{2.37} = -4.22$$

Step 4: Compare with Critical Value

- $Z = -4.22$ is **less than -1.96**, so we **reject** H_0.
- Conclusion: The average weight **is significantly different** from 500g.

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3. P-Value Approach

Concept

- The **p-value** represents the probability of obtaining a result **at least as extreme** as the observed value, assuming H_0 is true.
- If the **p-value is smaller than** α, we reject H_0.

Steps in the P-Value Method

1. **State the hypotheses** (H_0 and H_a).
2. **Choose** α (e.g., 0.05).
3. **Calculate the test statistic** (Z, t, or chi-square).
4. **Find the p-value from the statistical table**.
5. **Compare p-value with** α:
    - If **p-value** ≤ α → **Reject** H_0
    - If **p-value** > α → **Fail to reject** H_0

Example: P-Value Method

Using the same **weight test example**:

Step 1: Compute P-Value

From the **Z-table**, the probability for $Z = -4.22$ is **0.000013**.

Step 2: Compare with α

- **p-value = 0.000013**
- $\alpha = 0.05$
- Since **0.000013 < 0.05**, we **reject** H_0.

Conclusion:

The **p-value confirms the critical value decision**—the weight is significantly different from 500g.

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4. Key Differences: Critical Value vs. P-Value

Feature	Critical Value Approach	P-Value Approach
Definition	Uses a threshold (critical value) to decide rejection	Uses probability (p-value) to decide rejection
Comparison	Compares test statistic with critical value	Compares p-value with α
Decision Rule	Reject H_0 if test statistic > critical value	Reject H_0 if p-value < α
Output	Z, t, or chi-square critical values	p-value (probability)
Flexibility	Less flexible (fixed critical values)	More flexible (provides exact probability)
Usage	Used in traditional hypothesis testing	Used in modern statistical software

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5. Choosing Between Critical Value and P-Value

- **Use Critical Value** if:
  - You do not have access to statistical software.
  - You want a simple yes/no decision rule.
- **Use P-Value** if:
  - You need an exact probability value.
  - You use statistical software like Excel, SPSS, or Python.

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6. Summary & Final Thoughts

- **Critical Value Approach:** Compares test statistic with a **fixed threshold**.
- **P-Value Approach:** Uses **probability** to make a decision.
- **Both methods** lead to the **same conclusion**.